

QA REPORT

DriveNow Rentals

http://localhost:8102



Quality score · grade F

2 critical issues should be fixed before release.

42

pages explored

60

tests run

33

passed

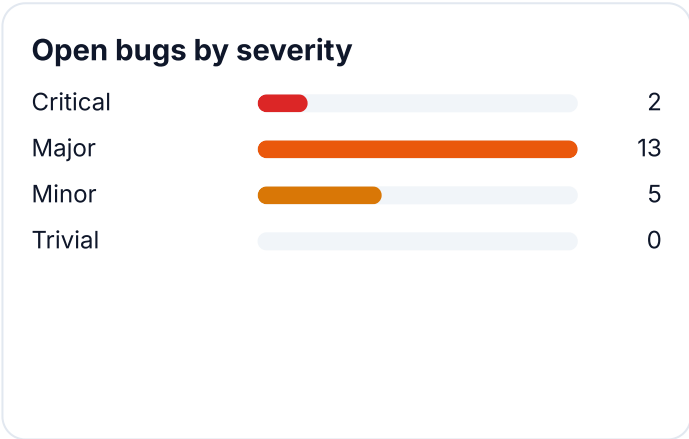
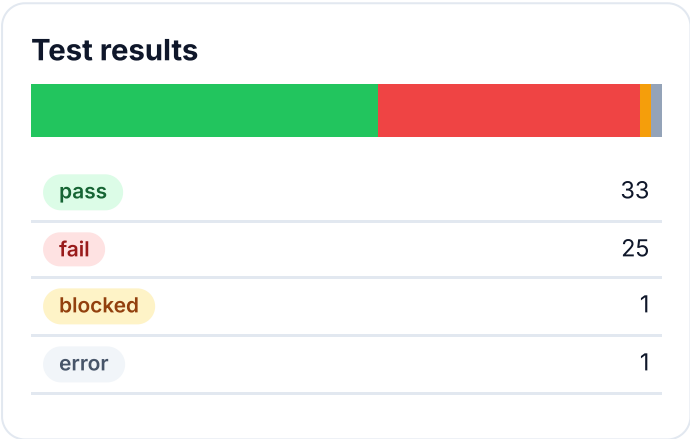
20

open bugs

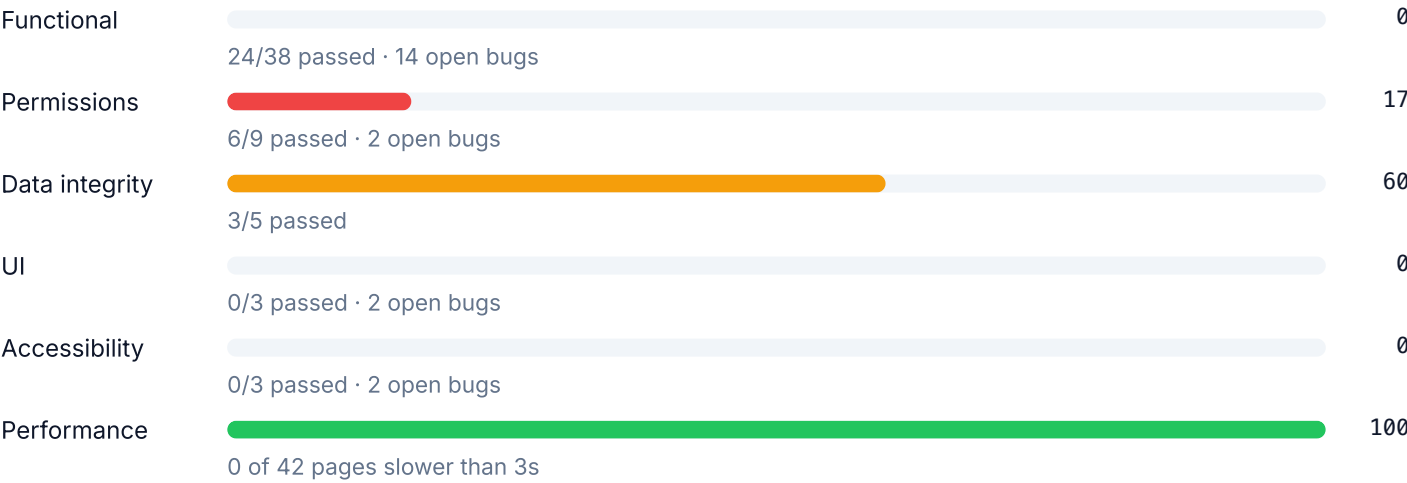
Executive summary

QA Pilot explored 42 pages of DriveNow Rentals as 3 signed-in roles (admin, agent, customer) and ran 60 tests: 33 passed and 25 failed. It reports 20 open bugs, of which 15 are critical or major.

- Critical: Agent can open restricted page /admin/revenue by direct URL (BUG-001).
- Critical: Customer can view another customer's booking via /bookings/{id} (IDOR) (BUG-002).
- Major: 7-day booking gets no 10% weekly discount; total \$840.00 instead of \$756.00 (BUG-003).
- Quality score 25.3/100 (grade F); weakest area: Functional (0).
- 2 permission holes: a role can open a screen it should not.
- 1 test could not run.
- 2 bugs need a person's review (flaky or low confidence).



Quality score breakdown



Formula: weighted mean of measured sub-scores (Functional 30, Permissions 20, Data integrity 20, UI 10, Accessibility 10, Performance 10); each = pass rate – open-bug penalties.

What QA Pilot found on the site

Rental · car rental back office (fleet, bookings, invoicing) (confidence 97%)

DriveNow Rentals is an internal car rental operations app. Staff manage the fleet, check car availability on a 14-day calendar, create and hand over bookings, record invoice payments and review revenue. Admins additionally add cars and see revenue reports.

ROLE	WHAT IT DOES
admin	Full access: manages fleet (add car, send to maintenance), views bookings, invoices, availability and the revenue report
agent	Creates bookings for customers, hands over cars, cancels bookings, views fleet, availability and invoices, records payments
customer	Signs in to browse fleet and availability, create bookings and view own bookings and invoices

Modules

Dashboard · Fleet · Availability · Bookings · Invoices · Revenue · Rental terms

Coverage heatmap

	SMOKE	CRUD	E2E	VALID.	BOUND.	EQUIV.	NEG.	STATE	PERM.	DATA	RULES	RESP.	A11Y
Access									1/1				
Accessibility													0/3
Availability										1/1			
Booking		2/2	1/1	2/2	1/2	2/4	0/1	4/6	2/3	0/1	2/5		
Car		1/1		1/1	0/1	1/2			2/2				
Dashboard										1/1			
Fleet			1/1	1/2			1/1			0/2			
Invoice			1/1					2/2	1/3		1/1		
Revenue										1/1			
Smoke	0/3												
UI												0/3	

Green: all passed · red: a test failed · amber: partly run or blocked · grey: not run · dashed: no tests (a gap).

Permissions

2 permission holes — screens a role can open although its own menus never lead there:

ROLE	SCREEN	EVIDENCE
agent	/admin/revenue	The page opened for agent: HTTP 200, title "Revenue · DriveNow Rentals", headings: Revenue report Collected revenue per car
customer	/admin/revenue	The page opened for customer: HTTP 200, title "Revenue · DriveNow Rentals", headings: Revenue report Collected revenue per car

Bugs (20)

BUG	TITLE	SEVERITY	PRIORITY	STATUS	REPRO
BUG-001	Agent can open restricted page /admin/revenue by direct URL	critical	P1	new	3/3
BUG-002	Customer can view another customer's booking via /bookings/{id} (IDOR)	critical	P1	new	3/3
BUG-003	7-day booking gets no 10% weekly discount; total \$840.00 instead of \$756.00	major	P2	new	3/3
BUG-004	Add car accepts negative Daily rate (-5) and saves car at \$-5.00	major	P2	new	3/3
BUG-005	Add car form accepts Daily rate of 0 and saves car at \$0.00	major	P2	new	3/3
BUG-006	Booking accepted for car in maintenance (Kia Picanto DN-106)	major	P2	new	3/3
BUG-007	Booking accepted when pickup date equals existing booking's return date	major	P2	needs review	1/3
BUG-008	Booking accepted with return date before pickup; negative days and total	major	P2	new	3/3
BUG-009	Booking estimate is \$48.00 instead of 96.00 for 2-day Golf booking (pickup 10-08, return 10-09)	major	P2	needs review	1/3
BUG-010	Broken link: page returns HTTP 404 — /terms-and-conditions	major	P2	new	seen on 3 page(s) / role(s)
BUG-011	Cancelled booking can be handed over; status changes to picked up	major	P2	new	3/3
BUG-012	Returned booking can be handed over again, status reverts to picked up	major	P2	new	2/3
BUG-013	Returned car still shown as "rented" on /cars after booking return	major	P2	new	3/3
BUG-014	Same-day rental priced as 0 days and \$0.00 instead of 1 day and \$42.00	major	P2	new	3/3
BUG-015	JavaScript error on /calendar	major	P2	new	3/3
BUG-016	/calendar overflows horizontally on small screens	minor	P3	new	3/3
BUG-017	/invoices overflows horizontally on small screens	minor	P3	new	3/3
BUG-018	Accessibility: Elements must meet minimum color contrast ratio thresholds on /calendar	minor	P3	new	3/3
BUG-019	Accessibility: Form elements must have labels on /bookings/new	minor	P3	new	3/3
BUG-020	Broken image — /cars/{id}	minor	P3	new	seen on 3 page(s) / role(s)

Agent can open restricted page /admin/revenue by direct URL

/admin/revenue is not in the agent menu, but typing its URL opens it with full content.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Revenue report

Grace Kim Agent Log out

\$210.00

COLLECTED

\$302.40

OUTSTANDING

Collected revenue per car

CAR	REVENUE
Toyota Corolla	\$210.00

BUG-001: expected Access is refused (403, redirect or error) for this role. — actual The page opened for agent: HTTP 200, title “Revenue · DriveNow Rentals”, headings: Revenue report | Collected...

Expected

Access is refused (403, redirect or error) for this role.

Actual

The page opened for agent: HTTP 200, title “Revenue · DriveNow Rentals”, headings: Revenue report | Collected revenue per car Also reproduced as customer.

Preconditions

- Logged in as agent

Steps to reproduce

1. Log in as agent
2. Open /admin/revenue by typing its URL in the address bar

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/admin/revenue
Role	agent
Found	2026-10-05 10:24 UTC
Reproducibility	3/3 · confidence 92%
Severity reason	Broken access control: a role reaches a screen meant for other roles that exposes privileged data or actions.
Linked tests	TC-INV-001, TC-INV-002
Requirements	—
Video	artifacts/bugs/BUG-001/clip.mp4

Customer can view another customer's booking via /bookings/{id} (IDOR)

Signed in as customer Liam O'Brien, opening /bookings/2 returns HTTP 200 and displays booking B-0002 belonging to Mei Tanaka, including her email, car, dates, price and invoice INV-0002. The booking list is correctly filtered to own bookings, but the detail URL has no ownership check. Reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0002

Liam O'BrienCustomerLog out

Booking

Customer

Mei Tanaka (mei@drivenow.test)

Car

Volkswagen Golf · DN-102

Pickup

2026-09-21

Return

2026-09-28

Status

returned

Price

Daily rate

\$48.00

Rental days

7

Subtotal

\$336.00

Weekly discount

-\$0.00

Total

\$336.00

Invoice INV-0002: \$302.40 (unpaid)

BUG-002: expected Opening another customer's /bookings/{id} is refused with 403/404 and no booking data is... — actual http://localhost:8102/bookings/2 returned HTTP 200 and showed booking B-0002 for Mei Tanaka...

Expected

Opening another customer's /bookings/{id} is refused with 403/404 and no booking data is shown.

Actual

http://localhost:8102/bookings/2 returned HTTP 200 and showed booking B-0002 for Mei Tanaka (mei@drivenow.test) with car, dates, price and invoice INV-0002, to customer Liam O'Brien.

Preconditions

- Customer A (Liam O'Brien) is signed in
- A booking belonging to another customer exists (B-0002, id 2, Mei Tanaka)

Steps to reproduce

1. Sign in as customer Liam O'Brien
2. Click "Bookings" (/bookings); verify only own bookings are listed (B-0010, B-0004, B-0007, B-0001)
3. Navigate directly to http://localhost:8102/bookings/2
4. Observe the page content and HTTP status

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/2
Role	customer
Found	2026-10-04 19:06 UTC
Reproducibility	3/3 · confidence 96%

Severity reason	Security/privacy failure: any customer can read other customers' personal data (name, email) and financial data (price, invoice) by changing the id in the URL.
Linked tests	TC-B00-027
Requirements	BR-011, US-011
Video	artifacts/bugs/BUG-002/clip.mp4

7-day booking gets no 10% weekly discount; total \$840.00 instead of \$756.00

A booking of Tesla Model 3 (\$120/day) from 2026-10-21 to 2026-10-28 is recorded as 7 days with subtotal \$840.00, but the weekly discount is -\$0.00 and the total is \$840.00. The 10% weekly discount should apply from exactly 7 days. The 6-day case behaved as expected (discount \$0.00, total \$720.00). The 8-day case was not reached because the test stopped at the 7-day failure.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0027

Grace KimAgentLog out

Booking confirmed. Estimated total \$840.00.

Booking

CustomerMei Tanaka (mei@drivenow.test)

CarTesla Model 3 · DN-104

Pickup2026-10-21

Return2026-10-28

Statusreserved

Hand over carCancel booking

Price

Daily rate\$120.00

Rental days7

Subtotal\$840.00

Weekly discount-\$0.00

Total\$840.00

BUG-003: expected 7 days: subtotal 840.00, discount 84.00 (10%), total 756.00. — actual Days 7, Subtotal \$840.00, Weekly discount -\$0.00, Total \$840.00. The confirmation message says 'Estimated...

Expected 7 days: subtotal 840.00, discount 84.00 (10%), total 756.00.	Actual Days 7, Subtotal \$840.00, Weekly discount -\$0.00, Total \$840.00. The confirmation message says 'Estimated total \$840.00'.
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Preconditions

- Tesla Model 3 (\$120.00/day) is available for 7 consecutive days in the future
- Logged in as agent
- Customer Mei Tanaka exists

Steps to reproduce

1. Open `http://localhost:8102/bookings/new`
2. Select Car: Tesla Model 3 — \$120.00/day
3. Select Customer: Mei Tanaka
4. Enter `pickup_date` 2026-10-21 and `return_date` 2026-10-28 (7 days)
5. Click Confirm booking
6. Observe the booking page (e.g. `/bookings/27`): subtotal, weekly discount and total

URL	http://localhost:8102/bookings/27
Role	agent
Found	2026-10-05 10:32 UTC
Reproducibility	3/3 · confidence 95%
Severity reason	Pricing is wrong: customers are charged \$84.00 too much on every 7-day booking, and there is no workaround on the booking screen. The error is a fixed 10% overcharge at the threshold boundary.
Linked tests	TC-B00-022
Requirements	BR-003, BR-004, BR-001, US-010, WF-01
Video	artifacts/bugs/BUG-003/clip.mp4

Console

http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined

Add car accepts negative Daily rate (-5) and saves car at \$-5.00

On the Add car form (/cars/new), entering -5 in 'Daily rate (\$)' is accepted. The app shows 'Car added to the fleet.' and the car detail page /cars/10 displays Daily rate \$-5.00. No validation message is shown. Reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

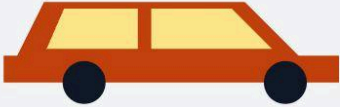
Invoices

Revenue

QAP_David Shaw QAP_William Heath

Omar Haddad Admin Log out

Car added to the fleet.



Plate

Category

Seats

Daily rate

Status

QAP-NJ-139X

Economy

3

\$-5.00

available

Send to maintenance

Booking history

BOOKING	CUSTOMER	PICKUP	RETURN	STATUS
No bookings yet.				

BUG-004: expected A daily rate of -5 is rejected with a validation message and nothing is saved. — actual The app shows 'Car added to the fleet.' and redirects to /cars/10, where Daily rate is listed as \$-5.00.

Expected

A daily rate of -5 is rejected with a validation message and nothing is saved.

Actual

The app shows 'Car added to the fleet.' and redirects to /cars/10, where Daily rate is listed as \$-5.00.

Preconditions

- Logged in as admin
- No existing car with the plate QAP-NJ-139X

Steps to reproduce

1. Open http://localhost:8102/cars/new
2. Enter Make: QAP_David Shaw
3. Enter Model: QAP_William Heath
4. Enter Plate: QAP-NJ-139X
5. Enter Seats: 3 (Category: Economy)
6. Enter Daily rate (\$): -5
7. Click 'Save car'
8. Observe the message and the resulting page

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/cars/10
Role	admin

Found	2026-10-05 10:36 UTC
Reproducibility	3/3 · confidence 72%
Severity reason	A negative daily rate is saved to the fleet, which is a money-related field. This can corrupt rental pricing and billing, and the invalid data is persisted without any error.
Linked tests	TC-CAR-003
Requirements	—
Video	artifacts/bugs/BUG-004/clip.mp4

Add car form accepts Daily rate of 0 and saves car at \$0.00

On the Add car form (/cars/new), entering a Daily rate of 0 is accepted. The car is saved with Daily rate \$0.00 and the success message "Car added to the fleet." is shown. Only positive rates should be accepted. Reproduced 3/3 times. Only the 0 value was exercised before the failure; -10 and 0.01 were not reached in the evidence.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Revenue

QAP Test

Omar HaddadAdminLog out

Car added to the fleet.



Plate

Category

Seats

Daily rate

Status

QAP-RATE0

Economy

5

\$0.00

available

Send to maintenance

Booking history

BOOKING	CUSTOMER	PICKUP	RETURN	STATUS
No bookings yet.				

BUG-005: expected The form rejects a Daily rate of 0 with a validation error. The car is not saved, and... — actual The car QAP-RATE0 is saved and the app redirects to http://localhost:8102/cars/14. The page shows Daily rate...

Expected

The form rejects a Daily rate of 0 with a validation error. The car is not saved, and only positive rates are accepted.

Actual

The car QAP-RATE0 is saved and the app redirects to http://localhost:8102/cars/14. The page shows Daily rate \$0.00 and the message "Car added to the fleet."

Preconditions

- Signed in as admin

Steps to reproduce

- Open http://localhost:8102/cars/new
- Enter Make: QAP
- Enter Model: Test
- Enter Plate: QAP-RATE0
- Enter Daily rate (\$): 0
- Click "Save car"

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/cars/14
Role	admin
Found	2026-10-05 10:38 UTC
Reproducibility	3/3 · confidence 78%

Severity reason	A car can be added to the fleet with a \$0.00 daily rate, which can lead to free or incorrectly priced rentals. There is no UI-level guard, and the bad record must be fixed manually after saving.
Linked tests	TC-FLE-003
Requirements	BR-018, US-002
Video	artifacts/bugs/BUG-005/clip.mp4

Booking accepted for car in maintenance (Kia Picanto DN-106)

An agent can create a booking for a car that admin sent to maintenance. The app confirmed the booking and created B-0010 with status reserved instead of refusing it. Reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0010

Grace KimAgentLog out

Booking confirmed. Estimated total \$35.00.

Booking

Customer

Mei Tanaka (mei@drivenow.test)

Car

Kia Picanto · DN-106

Pickup

2026-10-07

Return

2026-10-08

Status

reserved

Hand over car

Cancel booking

Price

Daily rate

\$35.00

Rental days

1

Subtotal

\$35.00

Weekly discount

-\$0.00

Total

\$35.00

BUG-006: expected Booking is refused with an error message and no booking is created for the maintenance... — actual
Redirected to http://localhost:8102/bookings/10 with message "Booking confirmed. Estimated total \$35.00."...

Expected

Booking is refused with an error message and no booking is created for the maintenance car.

Actual

Redirected to http://localhost:8102/bookings/10 with message "Booking confirmed. Estimated total \$35.00." Booking B-0010 was created for Kia Picanto DN-106 with status reserved.

Preconditions

- Logged in as agent
- Admin has sent one car (Kia Picanto DN-106) to maintenance

Steps to reproduce

1. Open http://localhost:8102/bookings/new
2. Select Car = "Kia Picanto — \$35.00/day" (the car in maintenance)
3. Select Customer = "Mei Tanaka"
4. Enter pickup_date = 2026-10-07 (today + 2 days)
5. Enter return_date = 2026-10-08 (today + 3 days)
6. Click "Confirm booking"

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/10
Role	agent
Found	2026-10-05 10:21 UTC
Reproducibility	3/3 · confidence 80%

Severity reason	A core booking rule is broken: a car out of service can be reserved, which risks customers being promised an unavailable vehicle. No workaround is apparent besides manual checking of car status by agents.
Linked tests	TC-B00-018
Requirements	BR-007, US-006, WF-03
Video	artifacts/bugs/BUG-006/clip.mp4

Console

```
http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined
http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined
```

Booking accepted when pickup date equals existing booking's return date

Creating a booking for Hyundai Tucson starting 2026-10-17 was confirmed even though an existing booking for the same car ends on 2026-10-17. The inclusive overlap check on boundary dates is not enforced. Reproduced in 1 of 3 runs.

DriveNow Rentals
Car rental operations

Booking B-0013 Grace Kim Agent Log out

Booking confirmed. Estimated total \$140.00.

Booking	
Customer	Mei Tanaka (mei@drivenow.test)
Car	Hyundai Tucson · DN-108
Pickup	2026-10-17
Return	2026-10-19
Status	reserved

Price	
Daily rate	\$70.00
Rental days	2
Subtotal	\$140.00
Weekly discount	-\$0.00
Total	\$140.00

BUG-007: expected The second booking is refused as overlapping, because pickup 2026-10-17 falls on the... — actual The booking was accepted. The app redirected to <http://localhost:8102/bookings/13> with the message "Booking..."

Expected

The second booking is refused as overlapping, because pickup 2026-10-17 falls on the existing booking's return date 2026-10-17 (inclusive overlap check).

Actual

The booking was accepted. The app redirected to <http://localhost:8102/bookings/13> with the message "Booking confirmed. Estimated total \$140.00." Booking B-0013 (Tucson, 2026-10-17 to 2026-10-19) now overlaps the existing 10-15 to 10-17 reservation. Reproducibility 1/3.

Preconditions

- Logged in as agent
- Hyundai Tucson has a reserved booking from 2026-10-15 to 2026-10-17 (customer Mei Tanaka)

Steps to reproduce

1. Open <http://localhost:8102/bookings/new>
2. Select Car "Hyundai Tucson — \$70.00/day"
3. Select Customer "Mei Tanaka"
4. Enter pickup_date 2026-10-15 and return_date 2026-10-17
5. Click "Confirm booking" (booking B-0012 is created)
6. Open </bookings/new> again
7. Select Car "Hyundai Tucson — \$70.00/day" and Customer "Mei Tanaka"

- 8. Enter pickup_date 2026-10-17 and return_date 2026-10-19
- 9. Click "Confirm booking"

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/13
Role	agent
Found	2026-10-05 10:22 UTC
Reproducibility	1/3 · confidence 47%
Severity reason	Double-booking the same car on a shared boundary date breaks a core booking rule and can leave the car committed to two customers, with no visible workaround for the agent. Reproduces only intermittently (1/3).
Linked tests	TC-B00-019
Requirements	BR-006, US-006, WF-01
Video	artifacts/bugs/BUG-007/clip.mp4

Console

```
http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined
http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined
```

Booking accepted with return date before pickup; negative days and total

On /bookings/new, submitting a return date one day before the pickup date is not rejected. Booking B-0017 was created and shown as confirmed with rental days -1 and total \$-42.00. The past-pickup validation does fire ("Pickup date cannot be in the past."), but no check exists for return earlier than pickup. Reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0017

Grace KimAgentLog out

Booking confirmed. Estimated total \$-42.00.

Booking

CustomerMei Tanaka (mei@drivenow.test)

CarToyota Corolla · DN-101

Pickup2030-06-15

Return2030-06-14

Statusreserved

Hand over carCancel booking

Price

Daily rate\$42.00

Rental days-1

Subtotal\$-42.00

Weekly discount-\$0.00

Total\$-42.00

BUG-008: expected Form is rejected with an error message that return date cannot be before pickup date, and... — actual Redirected to http://localhost:8102/bookings/17 with "Booking confirmed. Estimated total \$-42.00." Booking...

Expected Form is rejected with an error message that return date cannot be before pickup date, and no booking is created.	Actual Redirected to http://localhost:8102/bookings/17 with "Booking confirmed. Estimated total \$-42.00." Booking B-0017 was created with pickup 2030-06-15, return 2030-06-14, rental days -1, total \$-42.00.
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Preconditions

- Signed in as agent
- Cars and customers exist (Toyota Corolla at \$42.00/day, customer Mei Tanaka)

Steps to reproduce

- Open http://localhost:8102/bookings/new
- Select Car "Toyota Corolla — \$42.00/day"
- Select Customer "Mei Tanaka"
- Enter pickup_date 2030-06-15
- Enter return_date 2030-06-14
- Click "Confirm booking"

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/17

Role	agent
Found	2026-10-04 19:08 UTC
Reproducibility	3/3 · confidence 97%
Severity reason	A core workflow (booking creation) accepts invalid dates and persists a confirmed booking with a negative rental duration and a negative total (\$-42.00), which corrupts billing data. A workaround exists (agent enters correct dates), but the system does not prevent the error.
Linked tests	TC-B00-024, TC-B00-004, TC-B00-008
Requirements	BR-002, US-006, US-010, WF-01
Video	artifacts/bugs/BUG-008/clip.mp4

Booking estimate is \$48.00 instead of 96.00 for 2-day Golf booking (pickup 10-08, return 10-09)

Creating a booking for Volkswagen Golf (\$48.00/day) with pickup 2026-10-08 and return 2026-10-09 results in B-0029 with 1 rental day and an estimated total of \$48.00. The test expected 96.00 (2 days). Reproduced in 1 of 3 runs, and the test assumed today's date without confirming it.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0029

Grace KimAgentLog out

Booking confirmed. Estimated total \$48.00.

Booking

Customer

Liam O'Brien (liam@drivenow.test)

Car

Volkswagen Golf · DN-102

Pickup

2026-10-08

Return

2026-10-09

Status

reserved

Extend over car

Cancel booking

Price

Daily rate

\$48.00

Rental days

1

Subtotal

\$48.00

Weekly discount

-\$0.00

Total

\$48.00

BUG-009: expected Booking reserved with estimate 96.00 (2 days x \$48.00) for the dates from today+3 to... — actual Booking B-0029 reserved with 1 rental day and total \$48.00 (pickup 2026-10-08, return 2026-10-09). Message:...

Expected Booking reserved with estimate 96.00 (2 days x \$48.00) for the dates from today+3 to today+4, consistent across list, calendar, car history and dashboard.	Actual Booking B-0029 reserved with 1 rental day and total \$48.00 (pickup 2026-10-08, return 2026-10-09). Message: 'Booking confirmed. Estimated total \$48.00.'
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Preconditions

- Logged in as agent
- Volkswagen Golf is free for the requested dates

Steps to reproduce

1. Open http://localhost:8102/bookings/new
2. Select Car 'Volkswagen Golf — \$48.00/day'
3. Select Customer 'Liam O'Brien'
4. Enter pickup_date 2026-10-08
5. Enter return_date 2026-10-09
6. Click 'Confirm booking'
7. Observe redirect to /bookings/29 with message 'Booking confirmed. Estimated total \$48.00.'

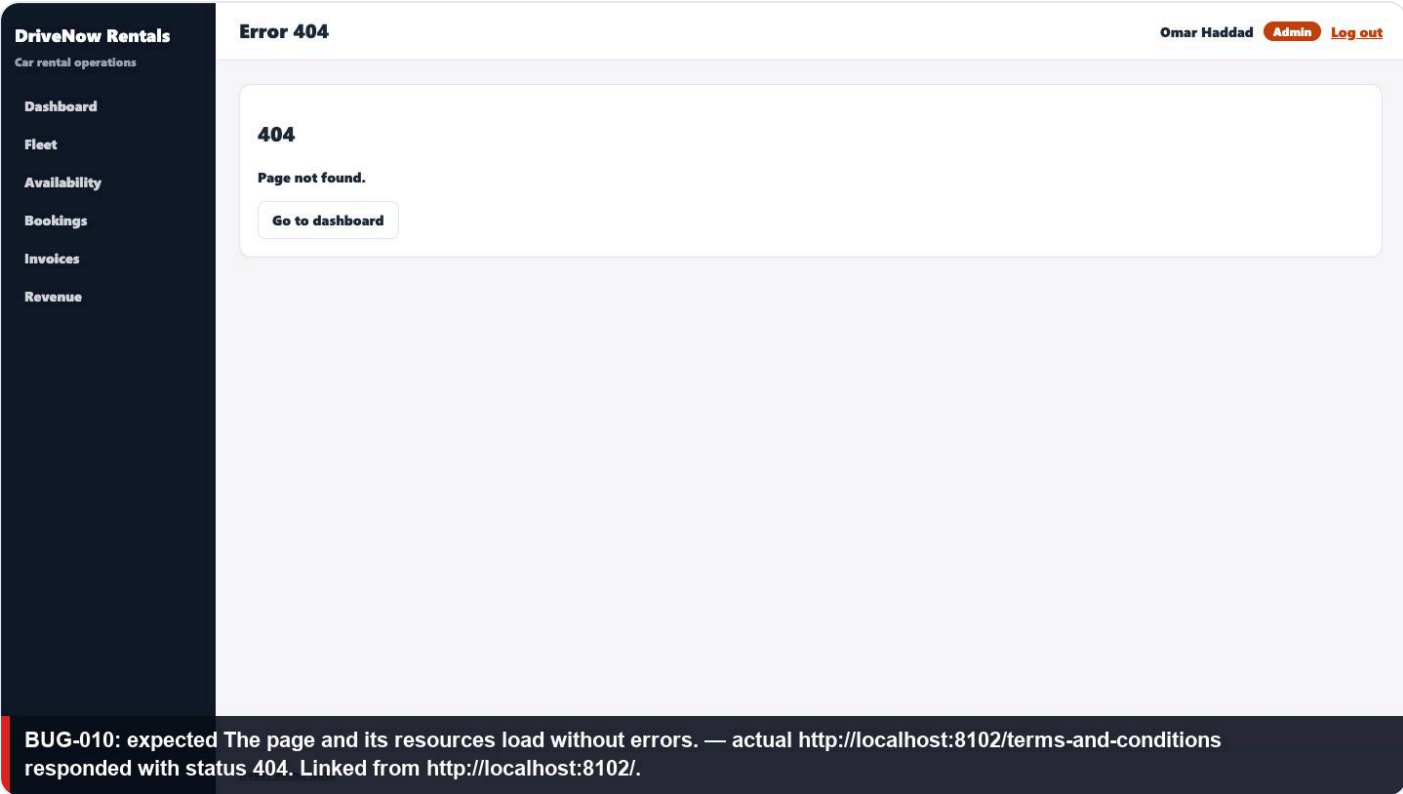
Role	agent
Found	2026-10-05 10:33 UTC
Reproducibility	1/3 · confidence 19%
Severity reason	Wrong rental price estimate on a core booking workflow affects money charged to customers; no workaround is visible on the form. Reproducibility is low (1/3) and the expected day-count rule is not confirmed, which limits certainty.
Linked tests	TC-B00-026
Requirements	BR-001, BR-004, US-006, US-009, US-001, WF-01
Video	artifacts/bugs/BUG-009/clip.mp4

Console

http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined
http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined
http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined

Broken link: page returns HTTP 404 — /terms-and-conditions

http://localhost:8102/terms-and-conditions responded with status 404. Linked from http://localhost:8102/.



Expected The page and its resources load without errors.	Actual http://localhost:8102/terms-and-conditions responded with status 404. Linked from http://localhost:8102/.
--	--

Steps to reproduce

1. Log in as admin
2. Open http://localhost:8102/terms-and-conditions

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/terms-and-conditions
Role	admin, agent, customer
Found	2026-10-05 10:53 UTC
Reproducibility	seen on 3 page(s) / role(s) · confidence 85%
Severity reason	Found by QA Pilot's automatic checks while exploring.
Linked tests	—
Requirements	—
Video	artifacts/bugs/BUG-010/clip.mp4

Cancelled booking can be handed over; status changes to picked up

On the booking detail page of cancelled booking B-0008, a "Hand over car" button is shown. Clicking it succeeds with "Car handed over to the customer." and the status changes from cancelled to picked up. The cancelled to picked up transition should be blocked. Reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0008

Grace KimAgentLog out

Booking

CustomerCarlos Mendes (carlos@drivenow.test)

CarVolkswagen Golf · DN-102

Pickup2026-10-13

Return2026-10-20

Statuscancelled

Hand over car

Price

Daily rate\$48.00

Rental days7

Subtotal\$336.00

Weekly discount-\$0.00

Total\$336.00

BUG-011: expected No "Hand over car" option is shown for the cancelled booking B-0008, and any hand over... — actual B-0008 shows a "Hand over car" button. Clicking it displays "Car handed over to the customer." and the status...

Expected

No "Hand over car" option is shown for the cancelled booking B-0008, and any hand over attempt is refused with the status staying cancelled.

Actual

B-0008 shows a "Hand over car" button. Clicking it displays "Car handed over to the customer." and the status changes to "picked up".

Preconditions

- Logged in as agent
- A booking with status cancelled exists (B-0008)

Steps to reproduce

1. Open http://localhost:8102/bookings
2. Click booking "B-0008" (href=/bookings/8), status is cancelled
3. Observe that a "Hand over car" button is displayed
4. Click "Hand over car"

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/8
Role	agent
Found	2026-10-04 19:03 UTC
Reproducibility	3/3 · confidence 83%
Severity reason	An invalid state transition on a core rental workflow: a cancelled booking can be handed over and become picked up, with no workaround in the UI and potential for releasing a car for a cancelled,

possibly unpaid booking.

Linked tests

TC-B00-016

Requirements

BR-009, US-007, WF-01

Video

artifacts/bugs/BUG-011/clip.mp4

Returned booking can be handed over again, status reverts to picked up

On the detail page of returned booking B-0002 (/bookings/2) a "Hand over car" button is available. Clicking it shows "Car handed over to the customer." and changes the status from returned to "picked up". The out-of-order transition returned → picked up is not blocked. Reproduced in 2 of 3 runs.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0002

Grace KimAgentLog out

Car handed over to the customer.

Booking

CustomerMei Tanaka (mei@drivenow.test)

CarVolkswagen Golf · DN-102

Pickup2026-09-21

Return2026-09-28

Statuspicked up

Return car

Price

Daily rate\$48.00

Rental days7

Subtotal\$336.00

Weekly discount-\$0.00

Total\$336.00

Invoice INV-0002: \$302.40 (unpaid)

BUG-012: expected Hand over on the returned booking is refused (button hidden or an error shown) and the... — actual The app showed "Car handed over to the customer." and B-0002's status changed to "picked up". Reproducibility...

Expected

Hand over on the returned booking is refused (button hidden or an error shown) and the status remains returned.

Actual

The app showed "Car handed over to the customer." and B-0002's status changed to "picked up". Reproducibility 2/3.

Preconditions

- Logged in as agent
- A returned booking exists (B-0002, id 2)
- A reserved booking exists (B-0011, id 11)

Steps to reproduce

1. Open <http://localhost:8102/bookings?status=returned> and confirm B-0002 is listed as returned
2. Open <http://localhost:8102/bookings/2>
3. Click the "Hand over car" button
4. Observe the message and the booking status

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/2
Role	agent
Found	2026-10-04 19:07 UTC

Reproducibility	2/3 · confidence 64%
Severity reason	An invalid lifecycle transition is accepted and a completed booking is reopened as picked up, corrupting booking status and car availability tracking. Agents can trigger it with one click; there is no obvious workaround besides manual correction.
Linked tests	TC-B00-017
Requirements	BR-009, US-007, WF-01
Video	artifacts/bugs/BUG-012/clip.mp4

Returned car still shown as "rented" on /cars after booking return

After clicking "Return car" on booking B-0009, the app shows "Car returned and invoice created", but /cars still lists Toyota Corolla as "rented" after reloading. The car status is not updated on return, so the car appears unavailable. Reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Fleet

Category All Filter

CAR	PLATE	CATEGORY	SEATS	DAILY RATE	STATUS
Toyota Corolla	DN-101	Economy	5	\$42.00/day	rented
Volkswagen Golf	DN-102	Compact	5	\$48.00/day	rented
Honda CR-V	DN-103	SUV	5	\$75.00/day	rented
Tesla Model 3	DN-104	Luxury	5	\$120.00/day	available
Ford Transit	DN-105	Van	9	\$95.00/day	available
Kia Picanto	DN-106	Economy	4	\$35.00/day	maintenance
BMW X5	DN-107	Luxury	5	\$160.00/day	available
Hyundai Tucson	DN-108	SUV	5	\$70.00/day	available
QAP David Shaw QAP William Heath	QAP-NJ-139	Economy	3	\$49.90/day	available
QAP David Shaw QAP William Heath	QAP-NJ-139X	Economy	3	\$-5.00/day	available
QAP David Shaw QAP William Heath	QAP-NJ-139S2	Economy	2	\$49.90/day	available
QAP David Shaw QAP William Heath	QAP-NJ-139S9	Economy	9	\$49.90/day	available
QAP Make QAP Model	QAP-001	Economy	5	\$60.00/day	available

Grace Kim

Agent

Log out

BUG-013: expected After the return, Toyota Corolla shows "available" on /cars and is free in the calendar. — actual Toyota Corolla still shows "rented" on /cars after the return of B-0009 and after reloading, despite the...

Expected

After the return, Toyota Corolla shows "available" on /cars and is free in the calendar.

Actual

Toyota Corolla still shows "rented" on /cars after the return of B-0009 and after reloading, despite the success message "Car returned and invoice created".

Preconditions

- Logged in as agent
- A picked up booking exists with the car rented (B-0009, Toyota Corolla)

Steps to reproduce

1. Open http://localhost:8102/cars and note Toyota Corolla is "rented"
2. Open http://localhost:8102/bookings
3. Click booking "B-0009" (/bookings/9)
4. Click the "Return car" button; message "Car returned and invoice created." appears
5. Open http://localhost:8102/bookings/9 to reload the booking
6. Open http://localhost:8102/cars and check the status of Toyota Corolla

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/cars
Role	agent
Found	2026-10-05 10:38 UTC
Reproducibility	3/3 · confidence 89%

Severity reason	Return completes and invoice is created, but the car stays "rented", so it appears unavailable for new bookings. This is an important fleet workflow with no evident workaround in the UI.
Linked tests	TC-FLE-006
Requirements	BR-010, US-009, WF-03, WF-01
Video	artifacts/bugs/BUG-013/clip.mp4

Same-day rental priced as 0 days and \$0.00 instead of 1 day and \$42.00

A booking with pickup and return on the same date is accepted but priced with 0 rental days. The confirmation message says 'Estimated total \$0.00', and the booking page shows Subtotal \$0.00 and Total \$0.00 although the daily rate is \$42.00. The expected minimum charge is 1 day. The failure reproduced 3/3.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Booking B-0011

Liam O'BrienCustomerLog out

Booking confirmed. Estimated total \$0.00.

Booking

Customer

Liam O'Brien (liam@drivenow.test)

Car

Toyota Corolla · DN-101

Pickup

2026-10-07

Return

2026-10-07

Status

reserved

Price

Daily rate

\$42.00

Rental days

0

Subtotal

\$0.00

Weekly discount

-\$0.00

Total

\$0.00

BUG-014: expected Booking is accepted with Rental days = 1 (max(1,0)) and an estimated total of \$42.00. — actual Booking B-0011 is accepted with the message 'Booking confirmed. Estimated total \$0.00.' The booking page...

Expected

Booking is accepted with Rental days = 1 (max(1,0)) and an estimated total of \$42.00.

Actual

Booking B-0011 is accepted with the message 'Booking confirmed. Estimated total \$0.00.' The booking page shows Rental days 0, Subtotal \$0.00, Total \$0.00 (daily rate \$42.00).

Preconditions

- Signed in as customer
- Toyota Corolla is available on today+2 (2026-10-07)

Steps to reproduce

- Open `http://localhost:8102/bookings/new`
- Select Car 'Toyota Corolla — \$42.00/day'
- Enter `pickup_date` 2026-10-07
- Enter `return_date` 2026-10-07 (same as pickup)
- Click 'Confirm booking'
- Observe the resulting page `http://localhost:8102/bookings/11` (booking B-0011)

Environment

Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)

URL

`http://localhost:8102/bookings/11`

Role	customer
Found	2026-10-05 10:22 UTC
Reproducibility	3/3 · confidence 95%
Severity reason	The core booking price calculation is wrong, so same-day rentals are charged \$0.00 instead of \$42.00. This is revenue loss, and the customer has no way to correct it.
Linked tests	TC-B00-021
Requirements	BR-001, BR-002, US-010, WF-01
Video	artifacts/bugs/BUG-014/clip.mp4

Console

http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined

JavaScript error on /calendar

JavaScript error: http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Revenue

Availability — next 14 days

Omar HaddadAdminLog out

freebookedmaintenance71 free car-days

Car	Mon 05	Tue 06	Wed 07	Thu 08	Fri 09	Sat 10	Sun 11	Mon 12	Tue 13	Wed 14	Thu 15	Fri 16	Sat 17	Sun 18
Toyota Corolla														
Volkswagen Golf														
Honda CR-V														
Tesla Model 3														
Ford Transit														
Kia Picanto														
BMW X5														
Hyundai Tucson														

BUG-015: expected The page loads without errors. — actual JavaScript error: http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined...

Expected

The page loads without errors.

Actual

JavaScript error: http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined Also reproduced as agent. Also reproduced as customer.

Steps to reproduce

- 1. Log in as admin
- 2. Open /calendar
- 3. Open the browser console

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/calendar
Role	admin
Found	2026-10-05 10:24 UTC
Reproducibility	3/3 · confidence 92%
Severity reason	Script error breaks part of the page's behaviour. Raised to major: the same symptom was also found by the automatic checks.
Linked tests	TC-SM0-001, TC-SM0-002, TC-SM0-003
Requirements	—
Video	artifacts/bugs/BUG-015/clip.mp4

Console

http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined

/calendar overflows horizontally on small screens

At 768px the page is wider than the screen, so users must scroll sideways.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Revenue

Availability — next 14 days

Omar HaddadAdminLog out

free

booked

maintenance

71 free car-days

Car	Mon 05	Tue 06	Wed 07	Thu 08	Fri 09	Sat 10	Sun 11	Mon 12	Tue
Toyota Corolla									
Volkswagen Golf									
Honda CR-V									
Tesla Model 3									
Ford Transit									
Kia Picanto									
BMW X5									
Hyundai Tucson									

BUG-016: expected The page fits the screen width without horizontal scrolling. — actual Content is 348px wider than the 768px screen (horizontal scrolling). Also reproduced as admin. Also... Reset demo data

Expected

The page fits the screen width without horizontal scrolling.

Actual

Content is 348px wider than the 768px screen (horizontal scrolling). Also reproduced as admin. Also reproduced as agent. Also reproduced as customer.

Steps to reproduce

1. Log in as admin
2. Open /calendar
3. Resize the window to 768px wide

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · tablet,mobile
URL	http://localhost:8102/calendar
Role	admin
Found	2026-10-05 10:25 UTC
Reproducibility	3/3 · confidence 92%
Severity reason	Layout defect on tablet/phone; content is still reachable by scrolling.
Linked tests	TC-UIX-001, TC-UIX-002, TC-UIX-003
Requirements	—
Video	artifacts/bugs/BUG-016/clip.mp4

Console

```
http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined
http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined
```

/invoices overflows horizontally on small screens

At 390px the page is wider than the screen, so users must scroll sideways.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Revenue

Invoices

Omar Haddad Admin Log out

INVOICE	DATE	CUSTOMER	DAYS	SUBTOTAL	DISCOUNT	TOTAL
INV-0001	2026-09-10	Liam O'Brien	5	\$210.00	\$0.00	\$210.00
INV-0002	2026-09-28	Mei Tanaka	7	\$336.00	\$33.60	\$302.40

BUG-017: expected The page fits the screen width without horizontal scrolling. — actual Content is

221px wider than the 390px screen (horizontal)

Expected

The page fits the screen width without horizontal scrolling.

Actual

Content is 221px wider than the 390px screen (horizontal scrolling). Also reproduced as agent. Also reproduced as customer.

Steps to reproduce

1. Log in as admin
2. Open /invoices
3. Resize the window to 390px wide

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · tablet,mobile
URL	http://localhost:8102/invoices
Role	admin
Found	2026-10-05 10:25 UTC
Reproducibility	3/3 · confidence 92%
Severity reason	Layout defect on tablet/phone; content is still reachable by scrolling.
Linked tests	TC-UIX-001, TC-UIX-002, TC-UIX-003
Requirements	—
Video	artifacts/bugs/BUG-017/clip.mp4

Console

```
http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined
http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined
```

Accessibility: Elements must meet minimum color contrast ratio thresholds on /calendar

axe-core reports a critical/serious WCAG violation on /calendar.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

Revenue

Availability — next 14 days

Omar HaddadAdminLog out

freebookedmaintenance77 free car-days

Car	Mon 05	Tue 06	Wed 07	Thu 08	Fri 09	Sat 10	Sun 11	Mon 12	Tue 13	Wed 14	Thu 15	Fri 16	Sat 17	Sun 18
Toyota Corolla														
Volkswagen Golf														
Honda CR-V														
Tesla Model 3														
Ford Transit														
Kia Picanto														
BMW X5														
Hyundai Tucson														

BUG-018: expected No critical or serious accessibility violations. — actual serious: Elements must meet minimum color contrast ratio thresholds (1×, e.g. .badge:nth-child(3)) Also... not demo data

Expected

No critical or serious accessibility violations.

Actual

serious: Elements must meet minimum color contrast ratio thresholds (1×, e.g. .badge:nth-child(3)) Also reproduced as agent. Also reproduced as customer.

Steps to reproduce

1. Log in as admin
2. Open /calendar
3. Run an axe-core / Lighthouse accessibility scan

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/calendar
Role	admin
Found	2026-10-04 19:01 UTC
Reproducibility	3/3 · confidence 92%
Severity reason	WCAG 2 A/AA violation that blocks or hinders assistive-technology users.
Linked tests	TC-ACC2-001, TC-ACC2-002, TC-ACC2-003
Requirements	—
Video	artifacts/bugs/BUG-018/clip.mp4

Console

http://localhost:8102/calendar 80:2 Uncaught ReferenceError: highlightToday is not defined

Accessibility: Form elements must have labels on /bookings/new

axe-core reports a critical/serious WCAG violation on /bookings/new.

DriveNow Rentals

Car rental operations

Dashboard

Fleet

Availability

Bookings

Invoices

New booking

Grace KimAgentLog out

Car

Select a car...

Customer

Select customer...

Pickup date

dd/mm/yyyy

Return date

dd/mm/yyyy

Rentals are charged per day. 7 days or more: 10% off. [Rental terms](#)

Confirm booking

BUG-019: expected No critical or serious accessibility violations. — actual critical: Form elements must have labels (2×, e.g. input[name="pickup_date"]) Also reproduced as customer.

Expected	Actual
No critical or serious accessibility violations.	critical: Form elements must have labels (2×, e.g. input[name="pickup_date"]) Also reproduced as customer.

Steps to reproduce

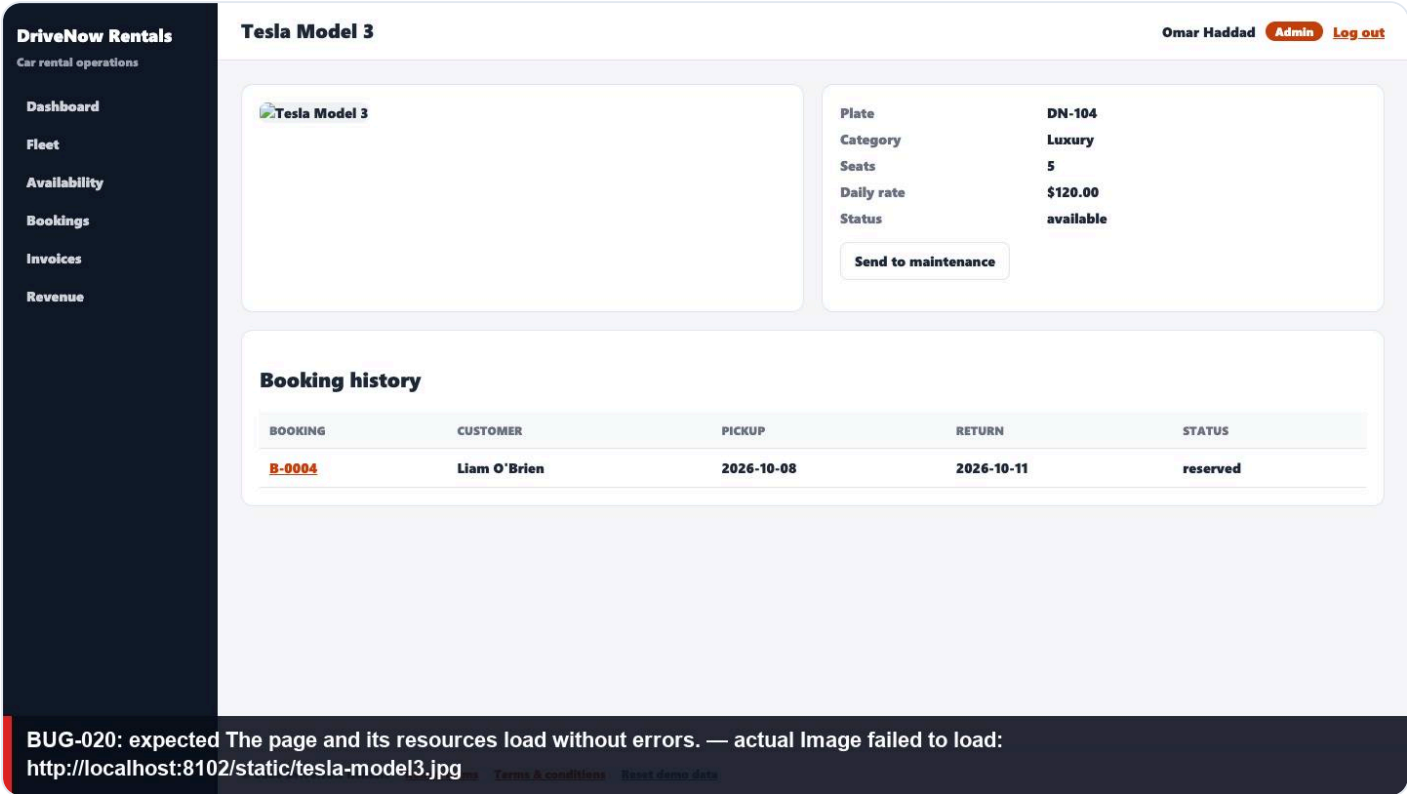
1. Log in as agent
2. Open /bookings/new
3. Run an axe-core / Lighthouse accessibility scan

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/bookings/new
Role	agent
Found	2026-10-04 19:01 UTC
Reproducibility	3/3 · confidence 92%
Severity reason	WCAG 2 A/AA violation that blocks or hinders assistive-technology users.
Linked tests	TC-ACC2-002, TC-ACC2-003
Requirements	—
Video	artifacts/bugs/BUG-019/clip.mp4

Console

http://localhost:8102/calendar 78:2 Uncaught ReferenceError: highlightToday is not defined

Image failed to load: http://localhost:8102/static/tesla-model3.jpg



Expected

The page and its resources load without errors.

Actual

Image failed to load: http://localhost:8102/static/tesla-model3.jpg

Steps to reproduce

- 1. Log in as admin
- 2. Open http://localhost:8102/cars/4

Environment	Chrome 154.0.8037.93 · Darwin 25.5.0 · 1440×900 (desktop)
URL	http://localhost:8102/cars/4
Role	admin, agent, customer
Found	2026-10-05 10:53 UTC
Reproducibility	seen on 3 page(s) / role(s) · confidence 85%
Severity reason	Found by QA Pilot's automatic checks while exploring.
Linked tests	—
Requirements	—
Video	artifacts/bugs/BUG-020/clip.mp4

Appendix A — Test results

TEST	TITLE	ROLE	RESULT	REPRO	BUGS
TC-ACC-001	Signed-out visitors cannot open protected screens	public	pass	—	
TC-ACC2-001	Accessibility scan of admin screens	admin	fail	3/3	BUG-018
TC-ACC2-002	Accessibility scan of agent screens	agent	fail	3/3	BUG-018, BUG-019
TC-ACC2-003	Accessibility scan of customer screens	customer	fail	3/3	BUG-018, BUG-019
TC-AVA-001	Maintenance car shown unavailable in calendar	admin	pass	—	
TC-B00-009	Admin cannot open /bookings/new by direct URL	admin	pass	—	
TC-B00-010	Admin cannot open /bookings/new?car={id} by direct URL	admin	pass	—	
TC-B00-011	E2E booking lifecycle: check calendar, reserve, hand over	agent	pass	—	
TC-B00-012	Hand over a reserved booking	agent	pass	—	
TC-B00-013	Cancel a reserved booking	agent	pass	—	
TC-B00-014	Cancel a picked up booking is refused	agent	pass	—	
TC-B00-016	Hand over a cancelled booking is refused	agent	fail	3/3	BUG-011
TC-B00-018	Maintenance car cannot be booked	agent	fail	3/3	BUG-006
TC-B00-019	Overlapping booking boundaries for the same car	agent	fail	1/3	BUG-007
TC-B00-021	Same-day rental counts as 1 day	customer	fail	3/3	BUG-014
TC-B00-022	Weekly discount threshold: 6, 7 and 8 days	agent	fail	3/3	BUG-003
TC-B00-027	Customer cannot open another customer's booking by URL	customer	fail	3/3	BUG-002
TC-B00-001	New booking: save with valid data	agent	pass	—	
TC-B00-005	New booking: save with valid data	agent	pass	—	
TC-B00-015	Cancel a returned booking is refused	agent	pass	—	
TC-B00-017	Reserved booking cannot jump to returned; returned cannot be handed over	agent	fail	2/3	BUG-012
TC-B00-020	Cancelled booking does not block the dates	agent	pass	—	
TC-B00-023	Estimate for a short rental with a non-round rate	customer	pass	—	
TC-B00-024	Return date before pickup is rejected	agent	fail	3/3	BUG-008
TC-B00-025	Pickup date in the past is rejected, today accepted	agent	pass	—	
TC-B00-026	New booking appears in list, calendar, car history and dashboard	agent	fail	1/3	BUG-009
TC-B00-002	New booking: required fields are enforced	agent	pass	—	
TC-B00-003	New booking: "pickup_date" rejects invalid input	agent	pass	—	
TC-B00-004	New booking: "return_date" rejects invalid input	agent	fail	3/3	BUG-008
TC-B00-006	New booking: required fields are enforced	agent	pass	—	

TC-B00-007	New booking: "pickup_date" rejects invalid input	agent	pass	—	
TC-B00-008	New booking: "return_date" rejects invalid input	agent	fail	3/3	BUG-008
TC-CAR-006	Agent cannot open /cars/new by direct URL	agent	pass	—	
TC-CAR-007	Customer cannot open /cars/new by direct URL	customer	pass	—	
TC-CAR-001	Add car: save with valid data	admin	pass	—	
TC-CAR-002	Add car: required fields are enforced	admin	pass	—	
TC-CAR-003	Add car: "Daily rate (\$)" rejects invalid input	admin	fail	3/3	BUG-004
TC-CAR-004	Add car: "Seats" boundaries	admin	error	—	
TC-CAR-005	Add car: "Seats" rejects invalid input	admin	pass	—	
TC-DAS-001	Cancelled booking leaves dashboard upcoming pickups	agent	pass	—	
TC-FLE-006	Returned car becomes available immediately	agent	fail	3/3	BUG-013
TC-FLE-001	E2E admin adds car, it becomes bookable, then goes to maintenance	admin	pass	—	
TC-FLE-002	Duplicate plate is rejected	admin	pass	—	
TC-FLE-003	Daily rate zero and negative rejected	admin	fail	3/3	BUG-005
TC-FLE-005	New car appears in fleet, filter and booking dropdown	admin	blocked	—	
TC-FLE-004	Seats outside 2-9 rejected server side	admin	pass	—	
TC-INV-001	Agent cannot open /admin/revenue by direct URL	agent	fail	3/3	BUG-001
TC-INV-002	Customer cannot open /admin/revenue by direct URL	customer	fail	3/3	BUG-001
TC-INV-003	E2E invoice payment reflected in revenue report	agent	pass	—	
TC-INV-004	Paying an already paid invoice is refused	agent	pass	—	
TC-INV-006	Invoice arithmetic: subtotal, discount, total	admin	pass	—	
TC-INV-007	Customer invoices list limited to own records and no Revenue menu	customer	pass	—	
TC-INV-005	Paid invoice cannot revert to unpaid	agent	pass	—	
TC-REV-001	Revenue counts only paid invoices	admin	pass	—	
TC-SM0-001	Every screen loads for admin	admin	fail	3/3	BUG-015
TC-SM0-002	Every screen loads for agent	agent	fail	3/3	BUG-015
TC-SM0-003	Every screen loads for customer	customer	fail	3/3	BUG-015
TC-UIX-001	Screens adapt to tablet and phone for admin	admin	fail	3/3	BUG-016, BUG-017
TC-UIX-002	Screens adapt to tablet and phone for agent	agent	fail	3/3	BUG-016, BUG-017
TC-UIX-003	Screens adapt to tablet and phone for customer	customer	fail	3/3	BUG-016, BUG-017

Appendix B — Traceability

Requirement → user story → test case → result → bug.

REQUIREMENT	STORIES	TEST	RESULT	BUGS
—	—	TC-ACC-001 Signed-out visitors cannot open protected screens	pass	
—	—	TC-ACC2-001 Accessibility scan of admin screens	fail	BUG-018
—	—	TC-ACC2-002 Accessibility scan of agent screens	fail	BUG-018, BUG-019
—	—	TC-ACC2-003 Accessibility scan of customer screens	fail	BUG-018, BUG-019
BR-007 Cars in maintenance cannot be booked	US-003, US-009	TC-AVA-001 Maintenance car shown unavailable in calendar	pass	
WF-03 Car status	US-003, US-009	TC-AVA-001 Maintenance car shown unavailable in calendar	pass	
—	—	TC-B00-009 Admin cannot open /bookings/new by direct URL	pass	
—	—	TC-B00-010 Admin cannot open /bookings/new?car={id} by direct URL	pass	
BR-001 Rental days equal return date minus pickup date, minimum 1 day	US-006, US-007, US-009	TC-B00-011 E2E booking lifecycle: check calendar, reserve, hand over	pass	
BR-009 Bookings progress reserved, picked up, returned in order	US-006, US-007, US-009	TC-B00-011 E2E booking lifecycle: check calendar, reserve, hand over	pass	
WF-01 Booking lifecycle	US-006, US-007, US-009	TC-B00-011 E2E booking lifecycle: check calendar, reserve, hand over	pass	
BR-009 Bookings progress reserved, picked up, returned in order	US-007	TC-B00-012 Hand over a reserved booking	pass	
WF-01 Booking lifecycle	US-007	TC-B00-012 Hand over a reserved booking	pass	
BR-008 Only reserved bookings can be cancelled	US-007	TC-B00-013 Cancel a reserved booking	pass	

BR-009 Bookings progress reserved, picked up, returned in order	US-007	TC-B00-013 Cancel a reserved booking	pass	
WF-01 Booking lifecycle	US-007	TC-B00-013 Cancel a reserved booking	pass	
BR-008 Only reserved bookings can be cancelled	US-007	TC-B00-014 Cancel a picked up booking is refused	pass	
BR-009 Bookings progress reserved, picked up, returned in order	US-007	TC-B00-014 Cancel a picked up booking is refused	pass	
WF-01 Booking lifecycle	US-007	TC-B00-014 Cancel a picked up booking is refused	pass	
BR-009 Bookings progress reserved, picked up, returned in order	US-007	TC-B00-016 Hand over a cancelled booking is refused	fail	BUG-011
WF-01 Booking lifecycle	US-007	TC-B00-016 Hand over a cancelled booking is refused	fail	BUG-011
BR-007 Cars in maintenance cannot be booked	US-006	TC-B00-018 Maintenance car cannot be booked	fail	BUG-006
WF-03 Car status	US-006	TC-B00-018 Maintenance car cannot be booked	fail	BUG-006
BR-006 A car cannot be booked for overlapping dates	US-006	TC-B00-019 Overlapping booking boundaries for the same car	fail	BUG-007
WF-01 Booking lifecycle	US-006	TC-B00-019 Overlapping booking boundaries for the same car	fail	BUG-007
BR-001 Rental days equal return date minus pickup date, minimum 1 day	US-010	TC-B00-021 Same-day rental counts as 1 day	fail	BUG-014
BR-002 Return date must be on or after pickup date	US-010	TC-B00-021 Same-day rental counts as 1 day	fail	BUG-014
WF-01 Booking lifecycle	US-010	TC-B00-021 Same-day rental counts as 1 day	fail	BUG-014
BR-003 Rentals of 7 days or more get a 10% weekly discount	US-010	TC-B00-022 Weekly discount threshold: 6, 7 and 8 days	fail	BUG-003

BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-010	TC-B00-022 Weekly discount threshold: 6, 7 and 8 days	fail	BUG-003
BR-001 Rental days equal return date minus pickup date, minimum 1 day	US-010	TC-B00-022 Weekly discount threshold: 6, 7 and 8 days	fail	BUG-003
WF-01 Booking lifecycle	US-010	TC-B00-022 Weekly discount threshold: 6, 7 and 8 days	fail	BUG-003
BR-011 Customers can only see their own bookings and invoices	US-011	TC-B00-027 Customer cannot open another customer's booking by URL	fail	BUG-002
—	—	TC-B00-001 New booking: save with valid data	pass	
—	—	TC-B00-005 New booking: save with valid data	pass	
BR-008 Only reserved bookings can be cancelled	US-007	TC-B00-015 Cancel a returned booking is refused	pass	
WF-01 Booking lifecycle	US-007	TC-B00-015 Cancel a returned booking is refused	pass	
BR-009 Bookings progress reserved, picked up, returned in order	US-007	TC-B00-017 Reserved booking cannot jump to returned; returned cannot be handed over	fail	BUG-012
WF-01 Booking lifecycle	US-007	TC-B00-017 Reserved booking cannot jump to returned; returned cannot be handed over	fail	BUG-012
BR-006 A car cannot be booked for overlapping dates	US-006	TC-B00-020 Cancelled booking does not block the dates	pass	
BR-008 Only reserved bookings can be cancelled	US-006	TC-B00-020 Cancelled booking does not block the dates	pass	
WF-01 Booking lifecycle	US-006	TC-B00-020 Cancelled booking does not block the dates	pass	
BR-001 Rental days equal return date minus pickup date, minimum 1 day	US-010	TC-B00-023 Estimate for a short rental with a non-round rate	pass	

BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-010	TC-B00-023 Estimate for a short rental with a non-round rate	pass	
BR-003 Rentals of 7 days or more get a 10% weekly discount	US-010	TC-B00-023 Estimate for a short rental with a non-round rate	pass	
WF-01 Booking lifecycle	US-010	TC-B00-023 Estimate for a short rental with a non-round rate	pass	
BR-002 Return date must be on or after pickup date	US-006, US-010	TC-B00-024 Return date before pickup is rejected	fail	BUG-008
WF-01 Booking lifecycle	US-006, US-010	TC-B00-024 Return date before pickup is rejected	fail	BUG-008
BR-016 Pickup date cannot be in the past	US-006	TC-B00-025 Pickup date in the past is rejected, today accepted	pass	
WF-01 Booking lifecycle	US-006	TC-B00-025 Pickup date in the past is rejected, today accepted	pass	
BR-001 Rental days equal return date minus pickup date, minimum 1 day	US-006, US-009, US-001	TC-B00-026 New booking appears in list, calendar, car history and dashboard	fail	BUG-009
BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-006, US-009, US-001	TC-B00-026 New booking appears in list, calendar, car history and dashboard	fail	BUG-009
WF-01 Booking lifecycle	US-006, US-009, US-001	TC-B00-026 New booking appears in list, calendar, car history and dashboard	fail	BUG-009
—	—	TC-B00-002 New booking: required fields are enforced	pass	
—	—	TC-B00-003 New booking: "pickup_date" rejects invalid input	pass	
—	—	TC-B00-004 New booking: "return_date" rejects invalid input	fail	BUG-008
—	—	TC-B00-006 New booking: required fields are enforced	pass	
—	—	TC-B00-007 New booking: "pickup_date" rejects invalid input	pass	

–	–	TC-B00-008 New booking: "return_date" rejects invalid input	fail	BUG-008
–	–	TC-CAR-006 Agent cannot open /cars/new by direct URL	pass	
–	–	TC-CAR-007 Customer cannot open /cars/new by direct URL	pass	
–	–	TC-CAR-001 Add car: save with valid data	pass	
–	–	TC-CAR-002 Add car: required fields are enforced	pass	
–	–	TC-CAR-003 Add car: "Daily rate (\$)" rejects invalid input	fail	BUG-004
–	–	TC-CAR-004 Add car: "Seats" boundaries	error	
–	–	TC-CAR-005 Add car: "Seats" rejects invalid input	pass	
BR-008 Only reserved bookings can be cancelled	US-001, US-007	TC-DAS-001 Cancelled booking leaves dashboard upcoming pickups	pass	
WF-01 Booking lifecycle	US-001, US-007	TC-DAS-001 Cancelled booking leaves dashboard upcoming pickups	pass	
BR-010 Returned cars become available immediately	US-009	TC-FLE-006 Returned car becomes available immediately	fail	BUG-013
WF-03 Car status	US-009	TC-FLE-006 Returned car becomes available immediately	fail	BUG-013
WF-01 Booking lifecycle	US-009	TC-FLE-006 Returned car becomes available immediately	fail	BUG-013
BR-017 Car plate must be unique and non-empty	US-002, US-003	TC-FLE-001 E2E admin adds car, it becomes bookable, then goes to maintenance	pass	
WF-03 Car status	US-002, US-003	TC-FLE-001 E2E admin adds car, it becomes bookable, then goes to maintenance	pass	
BR-017 Car plate must be unique and non-empty	US-002	TC-FLE-002 Duplicate plate is rejected	pass	

WF-03 Car status	US-002	TC-FLE-002 Duplicate plate is rejected	pass	
BR-018 Daily rate must be a positive number	US-002	TC-FLE-003 Daily rate zero and negative rejected	fail	BUG-005
WF-03 Car status	US-002, US-003	TC-FLE-005 New car appears in fleet, filter and booking dropdown	blocked	
BR-019 Seats must be between 2 and 9	US-002	TC-FLE-004 Seats outside 2-9 rejected server side	pass	
—	—	TC-INV-001 Agent cannot open /admin/revenue by direct URL	fail	BUG-001
—	—	TC-INV-002 Customer cannot open /admin/revenue by direct URL	fail	BUG-001
BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-008, US-004	TC-INV-003 E2E invoice payment reflected in revenue report	pass	
WF-02 Invoice payment	US-008, US-004	TC-INV-003 E2E invoice payment reflected in revenue report	pass	
BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-008	TC-INV-004 Paying an already paid invoice is refused	pass	
WF-02 Invoice payment	US-008	TC-INV-004 Paying an already paid invoice is refused	pass	
BR-003 Rentals of 7 days or more get a 10% weekly discount	US-005	TC-INV-006 Invoice arithmetic: subtotal, discount, total	pass	
BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-005	TC-INV-006 Invoice arithmetic: subtotal, discount, total	pass	
WF-02 Invoice payment	US-005	TC-INV-006 Invoice arithmetic: subtotal, discount, total	pass	
BR-011 Customers can only see their own bookings and invoices	US-011	TC-INV-007 Customer invoices list limited to own records and no Revenue menu	pass	
BR-012 Revenue report is restricted to administrators	US-011	TC-INV-007 Customer invoices list limited to own records and no Revenue menu	pass	

WF-02 Invoice payment	US-011	TC-INV-007 Customer invoices list limited to own records and no Revenue menu	pass	
WF-02 Invoice payment	US-008	TC-INV-005 Paid invoice cannot revert to unpaid	pass	
BR-004 Invoice subtotal is days times daily rate and total is subtotal minus discount	US-004	TC-REV-001 Revenue counts only paid invoices	pass	
WF-02 Invoice payment	US-004	TC-REV-001 Revenue counts only paid invoices	pass	
–	–	TC-SM0-001 Every screen loads for admin	fail	BUG-015
–	–	TC-SM0-002 Every screen loads for agent	fail	BUG-015
–	–	TC-SM0-003 Every screen loads for customer	fail	BUG-015
–	–	TC-UIX-001 Screens adapt to tablet and phone for admin	fail	BUG-016, BUG-017
–	–	TC-UIX-002 Screens adapt to tablet and phone for agent	fail	BUG-016, BUG-017
–	–	TC-UIX-003 Screens adapt to tablet and phone for customer	fail	BUG-016, BUG-017

Appendix C — How this report was made

QA Pilot explored the site in a real Chrome browser, built a model of its pages, roles and data, inferred requirements and business rules (confirmed by a person before testing), designed test cases with standard techniques, and ran them with an AI browser agent and rule-based checks. Every failure was re-run to measure reproducibility; flaky or low-confidence results are marked "needs review". Numbers in calculations were re-checked in Python where a business rule applied.

Safety: Full Mode on a test environment — test records are prefixed QAP_ and use fake data. Credentials are stored encrypted on the tester's computer and never appear in reports.

Authorization to test confirmed by Muzahidul Rahman on 2026-10-04.